

Daniel Han

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Data analyst specializing in behavioral analytics and statistical modeling. Designs data pipelines to uncover patterns in user behavior and sequential data. Published at EMNLP 2025.

PROFESSIONAL & LEADERSHIP EXPERIENCE

Johns Hopkins Center for Talented Youth (CTY) Baltimore, MD

Analytics Co-Lead | Data Analyst Jan 2025 - Present

- Built end-to-end data pipeline in Python (Pandas, NumPy, SciPy) to clean, transform, and analyze sequential event logs across 200+ multi-step puzzle sessions, engineering over 10 behavioral features from raw timestamped interactions, with automated exploratory analysis from ingestion through modeling.
- Developed regression framework using progress-based behavioral metrics, revealing that problem-solving strategies exist on a continuous spectrum rather than discrete clusters.
- Established that exploration-oriented strategies outperform narrow approaches ($r = -0.52$, $d = 1.14$), with distributed solvers showing ~3x less variability in outcomes. Preparing findings for pre-print.
- Deliver weekly research briefings to executive director, translating statistical results into recommendations for assessment design.
- Co-lead analytics team, defining methodological standards and directing research priorities across new task types.

Center for Language and Speech Processing Baltimore, MD

Graduate Researcher Oct 2024 - Feb 2025

- Co-authored **CLAIMCHECK** (EMNLP 2025), an annotated benchmark dataset for evaluating LLM performance on claim-centric peer review tasks across 60 NeurIPS submissions.

PROJECTS

Listening Drift: Music Behavior Analytics Dashboard | New York, NY Jan 2026 - Feb 2026

- Ingested ~3M listening events across 300+ users from Last.fm API into PostgreSQL, engineering daily behavioral features (entropy, mood proxies, temporal patterns) and 30-day rolling profiles.
- Applied PCA to 6-feature behavioral profiles mapping 303 users across intensity and diversity dimensions, with movement analysis revealing that volume and diversity are strongly correlated ($r = 0.67$) rather than opposing traits.
- Deployed interactive dashboard on Railway with animated population density maps, per-user behavioral trajectories, and configurable smoothing windows.

Hierarchical Bayesian Modeling of NFL Win Probability | Johns Hopkins University, Baltimore, MD Mar 2025 - May 2025

- Constructed hierarchical Bayesian models to decompose win probability into situational vs. team-level effects, applying partial pooling for stable estimates from sparse data (AUC ≈ 0.82).
- Quantified how context shapes performance (home-field advantage varies by team, $\sigma = 0.42$), showing that fixed-effect models underestimate team-level variation.

NarrativeNet: Character Relationship Extraction | Vanderbilt University, TN Oct 2023 - Dec 2023

- Engineered and deployed a Flask web application (**3rd Place, Vanderbilt AI Showcase**) for extracting character relationships from narrative text using NER and network centrality analysis.

EDUCATION

Johns Hopkins University, Whiting School of Engineering Baltimore, MD

MSE in Data Science | GPA: 3.74 | GRE: 162V, 170Q, 5W Aug 2024 - Dec 2025

Coursework: Deep Learning, Probabilistic ML, Bayesian Statistics, Natural Language Processing (NLP), Statistical Theory

Vanderbilt University Nashville, TN

BA in Mathematics, Minors in Data Science, Psychology, & Scientific Computing | Dean's List Aug 2020 - Dec 2023

Coursework: Mathematical Data Science, Machine Learning, Probability, Experimental Design

SKILLS & CERTIFICATIONS

Skills: Python (Pandas, NumPy, SciPy, NumPyro, PyTorch, scikit-learn), R, PostgreSQL, Tableau, Excel, Git, LaTeX

Languages: English (Native), Korean (Conversational)

Certifications: Carnegie Mellon CASOS Summer Institute, BCG Data Science & Advanced Analytics Virtual Program

Interests: Coffee, Poetry, Photography, Cognitive Psychology